KATHULAPALLI CHAKRADHAR REDDY

B.Tech in Computer Science & Engineering • Visvesvaraya National Institute of Technology (VNIT), Nagpur

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EDUCATION

Degree / Course	Institution / Board	CGPA / %	Year
B.Tech (CSE)	Visvesvaraya National Institute of Technology (VNIT), Nagpur	**7.02 / 10**	2022 – 2026
Class XII (BIEAP)	Narayana Junior College, Vijayawada	96.7%	2021
Class X (SSC)	Dr. KKR's Gowtham International School, Vijayawada	9.7 GPA	2019

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript (ES6+), TypeScript, SQL, Bash.

SDET & Quality Engineering: Pytest Framework, RFC 9111 HTTP Caching, dnspython (Anycast DNS), SQLAlchemy Testing, HTTPX Async Probing, Data Warehouse SQL Quality Gates, Performance SLA Benchmarking, Test Automation.

Backend & Cloud DevOps: Spring Boot, Spring Data JPA, FastAPI, RESTful APIs, JWT RBAC, Docker, Ansible, Terraform (IaC), AWS (EC2, VPC, S3), Netmiko, Cisco IOS (3750, 7200), GitHub Actions CI/CD.

AI & Reinforcement Learning: Multi-Agent RL (MARL), PyTorch, PettingZoo Parallel API, Ray RLlib, Actor-Critic Networks, SimPy Discrete Event Simulation, NetworkX, Temporal Anomaly Detection.

Databases & Core CS: PostgreSQL, MySQL, SQLite, MongoDB, Data Structures & Algorithms, OOP, Distributed Systems, Git, Postman.

FEATURED ENGINEERING PROJECTS

EdgeProbe: Automated CDN, DNS & Distributed Telemetry QA Framework [GitHub]

Python, Pytest, FastAPI, SQLAlchemy, dnspython, HTTPX, Docker Compose, SQLite/PostgreSQL

- Architected an automated SDET verification framework validating edge caching compliance (RFC 9111), Anycast DNS routing latency, and analytical telemetry ingestion integrity with **34 hermetic regression tests passing in 0.5s**.
- Asserted HTTP cache contracts including **X-Cache HIT/MISS transitions**, TTL expiration, conditional **304 Not Modified revalidation via ETags**, and upstream **Fast Purge** cache invalidation.
- Engineered multi-resolver DNS Anycast latency probers against Google (8.8.8.8) and Cloudflare (1.1.1.1), enforcing strict **sub-150ms SLA quality gates** using nanosecond-precision timers.
- Built automated SQLAlchemy assertions verifying telemetry warehouse ETL pipelines—detecting retry duplicates (`HAVING COUNT(*) > 1`), NULL constraint violations, negative latencies, and schema drift.

Zero-Poison: SW-MARL Defense for Cache Pollution in NDN [GitHub] • [Live Sim]

PyTorch, PettingZoo Parallel API, Ray RLlib, SimPy, NetworkX, Information-Centric Networking (NDN)

- Formulated a novel Sliding-Window Multi-Agent Reinforcement Learning framework preventing adversarial Cache Pollution Attacks (CPA) across distributed router nodes tracking an 11-dimensional temporal feature space.
- Designed an episode sliding window (W=20 to 200) with a threshold reset rule (P_window > 0 → Q → 0), achieving **0.0% poisoned content**, **92.3% cache hit rate**, and automatic self-healing without permanent router lockdown.
- Project lead under Prof. Nidhi Lal; research paper submitted to **ICICT 2026**.

Enterprise Network Automation & Real-Time Monitoring Platform [GitHub]

Python, FastAPI, Cisco IOS (Catalyst 3750, 7200), Ansible, Netmiko, Terraform, Docker, AWS

- Architected a multi-vendor automation and telemetry platform managing Cisco IOS routers & switches; automated Jinja2 template pushes, zero-touch provisioning, and daily Git-versioned configuration backups with unified diff tracking.
- Engineered an automated root-cause diagnostic engine for interface flaps, CRC errors, and OSPF adjacency drops; containerized backend via Docker and integrated Pytest test suites into GitHub Actions CI/CD.

Student Placement Management System [GitHub]

Java, Spring Boot, Spring Data JPA, React, MySQL, JWT, Spring Security, REST APIs

- Engineered a full-stack campus recruitment platform with RBAC (Student, Company, Admin) secured via JWT tokens and BCrypt encryption; enforced dynamic CGPA eligibility, duplicate prevention, and transactional cascade deletions.

LEADERSHIP & KEY ACHIEVEMENTS

- **Alumni Secretary**, Department of Computer Science & Engineering, VNIT Nagpur (2024–Present): Coordinated departmental alumni relations, student mentorship programs, and industry technical workshops.
- **Event Manager**, AXIS VNIT (Central India's Largest Annual Technical Festival): Led event operations, infrastructure logistics, budget allocation, and technical competition tracks.
- **Research Publication**: First author on research submission to **ICICT 2026** on Decentralized Multi-Agent RL Defense in Named Data Networks.